



# Automatic Speech Recognition System for Tonal Languages: State-of-the-Art Survey

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## Abstract

Natural language and human–machine interaction is a very much traversed as well as challenging research domain. However, the main objective is of getting the system that can communicate in well-organized manner with the human, regardless of operational environment. In this paper a systematic survey on Automatic Speech Recognition (ASR) for tonal languages spoken around the globe is carried out. The tonal languages of Asian, Indo-European and African continents are reviewed but the tonal languages of American and Austral-Asian are not reviewed. The most important part of this paper is to present the work done in the previous years on the ASR of Asian continent tonal languages like Chinese, Thai, Vietnamese, Mandarin, Mizo, Bodo and Indo-European continent tonal languages like Punjabi, Lithuanian, Swedish, Croatian and African continent tonal languages like Yoruba and Hausa. Finally, the synthesis analysis is explored based on the findings. Many issues and challenges related with tonal languages are discussed. It is observed that the lot of work have been done for the Asian continent tonal languages i.e. Chinese, Thai, Vietnamese, Mandarin but little work been reported for the Mizo, Bodo, Indo-European tonal languages like Punjabi, Latvian, Lithuanian as well for the African continental tonal languages i.e. Hausa and Yourba.

## 1 Introduction

Speech is observed as the valued as well as powerful innate method of communication among the people. Speech pass on the information related to speech production as well as perception without any specific skills or special tool. The speech signal provides the information regarding speech production environment (ENV), the medium by which it is transmitted as well as about speaker's characteristics. Every speaker has unique style of speaking as well physiological features of speech production. The speech signal contains the encoded information and it can be decoded effortlessly by the humans. So researchers have been intrigued by this

for developing the system that can automatically extract as well as process the large quantity of information from the speech. The wish of the researchers that human speech should be recognized by the machine, and the recognition should be either by the voice commands or by providing text result of speech. ASR system is the system in which the speech signals are converted into their matching text with the help of computer. It is the technology in which the computer maps the acoustic signals related to human speech into group of words or into actions. The recognized words are considered as final output and they are used as input in the work related to data entry, in controlling and commanding purpose and in preparing the document. Based on speech type, speaker model and based on vocabulary size, ASR system can be classified as given below.

- ASR based on speech
- ASR based on speakers
- ASR based on vocabulary size

Nowadays the ASR has become famous. Interaction between human and machine is growing fast in day to day applications as the computer science is developing. Many ASR has been developed with the purpose to make the daily routines easier for the person who is suffering from

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functional disabilities or is handicap. Methodical as well as comprehensive survey on ASR system for the tonal languages of the world has been put forward by this paper with the help of many key areas like dataset nature, study of various approaches and the evaluation parameters.

## 2 Motivations

The spoken word transcribed by computer into the readable text form is called ASR system. Various languages around the world are highly miscellaneous. To human, to culture as well as to environment the language is closely related. The miscellany of the language result in growth, development as well as in enrichment of civilization of humans. For maintaining cultural diversity in healthy way, the linguistic diversity is needed, as well as it become necessary to mankind's prosperity. Many languages have vanished as they are spoken by few speakers and due to their location in the rural or inaccessible areas of the globe. At present it is highly recommended that the technology must play an important role in stopping the extinction of the languages by providing the technologies of natural language processing as well as the ASR for all the tonal languages in the world.

In the world there are around 7000 languages, only for some of the languages ASR systems have developed. ASR system for languages is developed which is of large importance and has high economic value. By each passing day people are frequently shifted to the most dominant language. The acceptance of English or shifting towards English is the global trend. Most of the Asian, African, and Indo-European languages are also threatened by these unfavorable effects and appears to be at risk as endangered languages. Absence of efforts for developing the various tools for the speech processing for tonal languages for the African, Asian and Indo-European is observed. To have human-machine interaction more flexible, the ASR system for the tonal languages is required. This paper seeks to provide a systematized framework for the ideas that have been discussed above.

## 3 Background

In this section, the speech and its classification, tone and its features, tonal language and its phonematic structure, classification of tonal languages, approaches, Speech Recognition (SR) issues, objectives, applications are discussed.

### 3.1 Speech and Its Classification

Speech is the exchange of spoken words or the capability of expressing thoughts as well as feelings with articulate

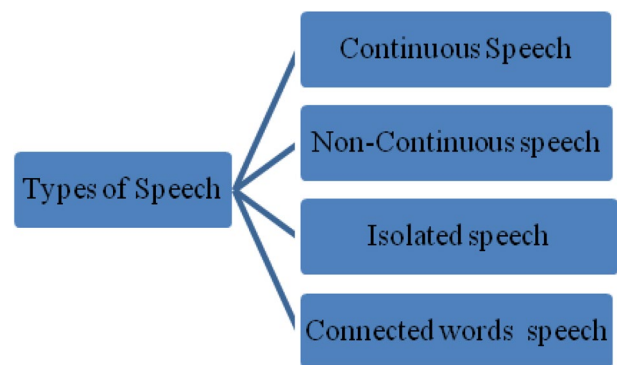


Fig. 1 Types of speech

sounds. Speech has many types of classifications. Figure 1 illustrates the forms of speech which are used commonly.

The continuous speech is the speech, which is spoken naturally, and all the spoken words are correctly and accurately identified by the speech recognition system. Non-continuous speech, also known as spontaneous speech, is the speech in which the speaker speaks single words and each of these single words are accurately identified by the speech recognition system. The isolated speech is the speech where an individual word or an individual utterance is considered at the sampling window time. In connected words speech separate utterance is run together and minimum pause is observed between them.

### 3.2 Tone and Tonal Language

In learning system as well as in speech recognition, tone recognition plays a vital role. For each syllable of language when there is change in pitch is called tone. The language is said to be a tonal language, if the meaning of the word is changed with the pitch of word. The vowels in the tonal language must be recognized first. The Fundamental Frequency ( $f_0$ ) act as a cue for tone and phoneme acts as a cue for first, second formants ( $f_1$ ,  $f_2$ ). Morphemes are distinguished by the pitch variation. ASR of tonal language needs the detection of tone, consonants and vowels of syllable. Tonal features are given below.

- $f_0$  and its  $f_0$  difference ( $\Delta f_0$ )
- Subsection of  $f_0$  and  $f_0$  intercept
- Subsection mean of  $f_0$  and mean of  $\Delta f_0$

### 3.3 Tonal and Non-tonal Languages Classification

Tonal and non-tonal language classification system can be classified on the basis of pitch level and speed of pitch. Wang et al. [131] proposed a system that classifies the language

as tonal and non tonal language. The system consists of the following components

- *The pitch extraction module:* In this module the extraction of raw f0 from the input speech is done on the basis of autocorrelation method.
- *The pitch smoothing and trimming module:* In this module the trimming algorithm is used for comparing the Average Pitch Values of Voiced Segment ( $f_v$ ) with Average Pitch Value of Whole Utterance (F). It removes sharp spikes in the pitch tracing.
- *The pitch changing speed analysis module:* In this module the calculation of absolute value of pitch variation is done within voiced segment.
- *The pitch changing level analysis module:* The pitch change analyzed is performed for each voiced segment in the pitch changing level analysis module.
- *The voiced duration counter module:* Total number of voice segments counted in the speech give the voice duration in the voiced duration counter module
- *The pitch average value calculation module:* The pitch average is calculated for normalizing the speed as well as the level of pitch change between various genders in the pitch average value calculation module.
- *The voiced phoneme counter module:* The voiced phoneme is identified with the help of broad-phone class recognizer in the voiced phoneme counter module.
- *The feature parameters normalization module:* The feature parameters normalization module helps in the final classification.
- *The classifier:* Different type of classifier is used to evaluate the system performance and it helps to classify the language as tonal and non-tonal language.

The block diagram for the tonal and non-tonal language classification system is given in Fig. 2.

### 3.4 Classification of Tonal Languages

Tonal languages can be classified based on the language family or based on the continents. For understanding and for the simplification we classify tonal languages based on the continents (Table 1).

The phonological features of the tonal languages are given in the Table 2

### 3.5 Approaches to ASR System

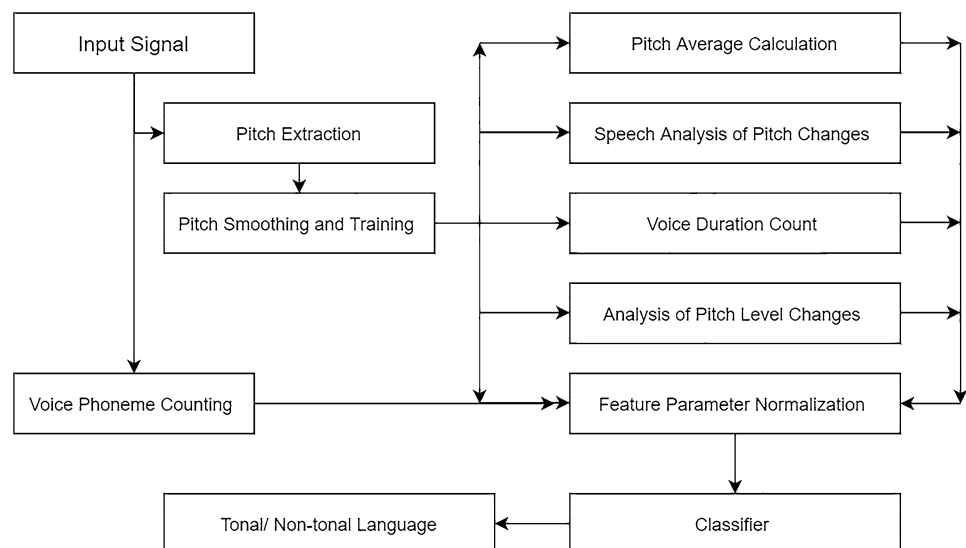
The main purpose of ASR system is to compare the various speech signals with respect to various phonemes, syllables, words/sentences. To construct ASR system there are three techniques can be adopted which are

- Acoustic Phonetic Approach
- Pattern Recognition Approach
- Artificial Intelligence (AI) Approach

#### 3.5.1 Acoustic Phonetic Approach

It is the earliest approach of SR. The acoustic phonetic approach is based on evaluation of the speech sounds as well as providing these sounds with appropriate labels. The acoustic approach consist of the phonemes that are finite, distinctive and are characterized by acoustic properties that are embedded over time in the speech signal. The acoustic approach is straightforward and it can be learned with the machine.

**Fig. 2** Block diagram of tonal and non-tonal language classification system (Wang et al. [131])



**Table 1** Classification of some prominent tonal languages

| Continents    | Tonal languages                                                                                           |
|---------------|-----------------------------------------------------------------------------------------------------------|
| African       | Yoruba, Hausa, Kanuri, Songhay, Maninka, Sandawe, Nubain, Dinka, Luo, Ewe, Lingala                        |
| American      | Navajo, Tzotzil, Cherokee                                                                                 |
| AustralAsia   | Cemuhi, Siane                                                                                             |
| Asia          | Mandarin, Chinese, Vietnamese, Thai, Laotian, Bodo, Mizo                                                  |
| Indo-European | Punjabi, Latvian, Lithuanian, Dogri, Lahnda, Serbo-Croatian, Saramaccan Slovene, Luxemburgish, Papiamento |

**Table 2** Phonological features of tonal languages

| Language name | Classification family | Total tones | Types of tones                                             | Vowels | Consonants |
|---------------|-----------------------|-------------|------------------------------------------------------------|--------|------------|
| Chinese       | Sino-Tibetan          | 6           | High, highrising, lowfalling-rising, high falling, neutral | 5      | 19         |
| Mandrain      | Sino-Tibetan          | 5           | High, rising, falling-rising, falling                      | 7      | 22         |
| Thai          | Tai-Kadai             | 5           | Mid, low, falling, high, rising                            | 18     | 20         |
| Vietnamese    | Austro-Asiatic        | 6           | Level, falling, broken, curve, rising, drop                | 11     | 25         |
| Mizo          | Sino-Tibetan          | 4           | High, low, rising, falling                                 | 5      | –          |
| Bodo          | Tibeto-Burman         | 2           | High, low                                                  | 6      | 16         |
| Yoruba        | Niger-congo           | 3           | High, mid, low                                             | 11     | 17         |
| Hausa         | Afro-Asiatic          | 2           | High, low                                                  | 10     | 33         |
| Lithuanian    | Indo-European         | 2           | Rising, falling                                            | 10     | 37         |
| Latvian       | Indo-European         | 3           | Even, broken falling                                       | 11     | 25         |
| Swedish       | Indo-European         | 2           | Single, double                                             | 17     | 19         |
| Punjabi       | Indo-European         | 3           | High, mid, low                                             | 10     | 25         |

### 3.5.2 Pattern Recognition Approach

In pattern approach, mathematical formulated framework is used, and consistent speech pattern representations are established for performing reliable pattern comparison with the labeled training samples through training algorithm.

### 3.5.3 Artificial Intelligence Approach

Artificial intelligence approach is the combination of acoustic phonetic as well as pattern recognition approach. The AI approach in the SR compile and add knowledge from various sources. The recognition procedures is mechanize according to the method by which the person applies his/her intelligence to visualization, analysis and makes the final decision on the basis of acoustic features that are measured.

## 3.6 Issues in ASR

The basic problem in the SR system is the pattern matching. The main purpose is to take the pattern and the speech signal and classify it as the last learned pattern in a sequence. The performance as well as the reliability of ASR is affected by the following reasons.

*Speakers:* Every human has unique voice and hence developing techniques that can recognize every human voice

with accurately as well as recognize the dialect of the language is a major issue.

*Speaking Rate and Style:* Humans speak with various rates and have dissimilar pronunciations for similar words, therefore it become difficult to have stable patterns for the sound or for the words for every person and with their style of speaking.

*Co-articulation:* The spoken words spectral feature varies and depends on the words that surround it.

*Absence of the Speech Model:* Absence of sophisticated as well traceable speech model is noticed.

*ENV:* In the real world situations the speech signal is always associated with the noise, e.g. vehicle noise, television. It is difficult to recognize the speech when spoken in the noisy environment.

Each of the above-mentioned factors becomes the major issue in the ASR system as they contribute in the variability of the speech signals. These issues must be considered while developing the ASR system.

## 3.7 Objectives of ASR System

The ASR system must be successfully utilized in the real-world conditions. The accuracy of the ASR system must be high so that there is hardly any mismatch between the

training and the testing phase speech. The various objectives of the ASR are:

- To have the techniques that can extract the important speech features for achieving the robustness in the ASR system.
- To enhance already existing features extraction techniques for the accuracy and efficiency.
- To have high accuracy as well as recognition rate at various stationary as well as non-stationary noises.
- To know SR and the way the SR work.
- To view SR application in various areas.
- To know SR implementation as a single application.
- To check how SR is faster than writing by hand and checking hand free capability of SR
- To make SR useful for mental and physical disabled person
- To use SR for the various application like voice dialing industry, dictation and navigation.

### 3.8 Applications Areas of ASR System

SR has enabled the speech technology to be used in various applications. The major field in which the SR is being used is listed below

- Mobile phones
- Automatic vehicles
- Industrial devices
- Military devices
- Communication
- Educational filed
- Medical and health care

## 4 Survey Protocol

The purpose of survey protocol is to tell about the comprehensive layout plan related to tasks such as inventory and monitoring. The set of rules as well as guidelines are also depicted by protocol for performing the survey on the literature work related to tonal languages ASR. It helps the novice researchers in this field with enough details regarding the nature of the ASR system for the tonal languages of the world. The survey protocol states the sub concepts like planning, research questions, sources of study, quality assessment, and sources of dataset, inclusion as well as exclusion criteria and the data extraction methods. This survey also help the user regarding the requirement of the system under the study, how the study is done and where the study is helpful. This survey also provides the guidance related to cost, methods related to data collection, management, analysis as well as reporting of the outputs.

### 4.1 Planning the Survey

With the purpose as well as the reason for the studying the ASR system for tonal languages, begins with the planning of the survey. The techniques related to approach the target; inclusion as well as exclusion criteria.

### 4.2 Research Questions

The main purpose of the formulation of research questions in this systematic survey review is to help the readers to know what this systematic review intends to answer. The questions are selected in such a manner that the area of SR is totally covered. In this systematic review questions related to SR and tone languages are addressed in the Table 3 which depicts the specific research questions and sub-questions.

### 4.3 Speech Corpus Referred

- Large Vocabulary Thai Continuous Speech Corpus (LOTUS)
- Vietnamese Broadcast News Corpus (VNBN)
- Voice of Vietnamese (VOV)
- Open Linguistic Resources Channeled Toward Interdisciplinary Research (ORCHID)
- National Electronics and Computer Technology Center (NECTEC)
- National Electronics and Computer Technology Center Advanced Telecommunications Research (NECTEC-ATR)
- Credit Card Corpus
- DARPA RM SI
- DARPA RM
- TIMIT
- TIDigits
- TINIST Corpus
- VNSpeech Corpus

### 4.4 Inclusion and Exclusion Criteria

Only papers related to speech recognition and tone are selected. The classifications of the papers are done on the basis of years, dataset, feature extraction techniques and evaluation measure. The papers published in journals and proceedings of the flagship conferences are included. The papers other than speech recognition and tone are excluded.

### 4.5 Quality Assessment

After the inclusion/exclusion criterion is used to find out the relevant paper the quality assessment is performed on the remaining papers through quality assessment form

**Table 3** Research questions and sub-questions

| Research Questions                                                                                   | Motivations                                                                                                                                                         |
|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What is ASR?                                                                                         | To have knowledge about the approaches suitable for languages                                                                                                       |
| What are the various approaches to build ASR system for tonal languages?                             | To find and study methods as well as literature work for ASR system for tonal languages                                                                             |
| Number of tonal languages in which ASR system is successful?                                         | To study that for how many tonal languages ASR system has been successfully developed                                                                               |
| What feature extraction methods and classifiers will be the best for ASR system for tonal languages? | To explore the methods for ASR for tonal languages. It urges to study the various features and classifiers helpful for the study                                    |
| Work done in Asian, African, Indo-European, Austro-Asia is?                                          | It motivates the user for the comparative study for Asian, African, Indo-European, Austro-Asia tonal languages and reports the reviewed work                        |
| Applications where ASR is most suited?                                                               | To know advantages in a real time world to make ASR system more demanding                                                                                           |
| Datasets availability for the research work?                                                         | Helps in studying the enough availability of datasets for the tonal languages                                                                                       |
| Is the datasets size affect the accuracy of the system proposed?                                     | Practical experiments to be performed to know and compare the accuracies of identifying the speech, words and sentences with the selected algorithms and techniques |
| Is the accuracy with high rate is achieved with the hybrid approach features?                        | To study the experiments with hybrid approach and analyze the comparison on the accuracy rates                                                                      |

(Appendix 1). All the papers included in the review contain the high-quality speech recognition research, the questions included in the Section 1 of the Appendix 1 serve as basis for screening the study. After the research paper is included, the study is done for the classification based on Sect. 2, then preceded to Sects. 3 and 4. The study incorporates

- Research papers of high quality
- Datasets and speech corpus of high quality.
- Comprehensive survey
- Topic confined
- Approaches related to SR inculcated
- Enough data for conducting comparison among various studies
- Mentioned the forms, methods for SR system

#### 4.6 Data Extraction

During the starting of the review, many problems were faced. Extraction of all the relevant data (Appendix 2) from many studies was very difficult. Due to this problem, it was necessary to contact various researchers, to find the needed details which were not possibly inferred from the research paper. The steps carried in the data extraction are as follows

- All the papers were surveyed firstly and then from the primary studies data has extracted.
- Data extraction consistency was checked by another researcher by performing the data extraction on primary data.
- The samples were randomly selected, and the cross checking of the results was done.

- During the cross checking if any disagreement was noticed it was resolved.

The following data items were studied in the data extraction phase.

- Papers were marked with an identity number which formulates the unique id.
- Bibliographic data was collected about the author, year, title and the source.
- The name of the journal or the conference proceedings were mentioned.
- Objective of the research.
- Feature extraction, classification techniques used, and accuracy achieved.
- Identification rate and error rates obtained after the experiments.
- Careful examination of the survey or state of art was presented by the authors.
- Future recommendations were made.

## 5 Reported Work on Asian Tonal Languages

### 5.1 Chinese

Fu et al. [20] studied that, primarily, temporal envelope cues provided high level performance of Mandarin Chinese SR, by extending the results previously obtained with English language based on tonal nature. The authors studied differences between Chinese and English SR. The correct level achieved for tones was, 80% at 500 Hz envelope

filter condition for the recognition. 11.0% and 2.9% were produced as significant difference in open set recognition between Chinese and English sentences, respectively. Jiyong et al. [36] proposed the acoustic approach called the Intra Syllable Dependent Phone Set (ISDPs) where Speech Recognition Unit (SRU) was taken as ISDPs for the Chinese SR. The ISDPs gave 93.29% accuracy rate whereas Mixed Gaussian Continuous Probability Model (MGCPM) gave 93.78%. Zhang and Hirose [142] proposed approach on the recognition of Chinese lexical tone. The new approach stated that the preceding lexical tone offset and the succeeding lexical tone onset act as anchor points of the pitch heights of onset as well as offset lexical tone. The new tone recognition method was developed by studying the physical parameter that is the Time Periods of the Successive Human Vocal Cords Vibrations (F0) measured in Hertz and relative heights between neighboring tones. The full syllable method obtained 75.3% tone recognition accuracy whereas the proposed method accounted 85.5% tone recognition accuracy. Wang [130] performed experiment for syllable indexing for retrieval of Mandarin Chinese spoken words with the help of Vector Space Model (VSM) and Syllable Lattice Matching (SLM) approach as well as with combined approach (VSM + SLM). SLM perform better than VSM. Li and Sun [136] proposed State Feedback Neural Network Activation Model (SFNNAM) which was based on the characteristics of Chinese speech structure. Less calculation was needed in SFNNAM model as compared to Hidden Markov Model (HMM). The SFNNAM gave 93.5% of recognition accuracy for Chinese number. Zhang et al. [140] build the Context Dependent (CD)-Initial/Final (IF) acoustic model which used the concept of decision tree based on the state type approach as well as adaptive mixture increasing strategy. The Extended Initial/Final (XIF) set was based on Chinese language characteristics. Syllable Accuracy (SA) obtained by CD-XIF model was 72.21% and it was 58.67% by Context Independent (CI)-XIF model. Tsai and Chang [124] build the dialect identification system that was capable of recognizing the Chinese dialects, Holo, Hakka and Mandarin by using the Gaussian Mixture Bigram Model (GMBM). The discriminative training of GMBM based on Chinese dialect identification system employed with Minimum Classification Error (MCE) for greater efficiency. Zhang et al. [143] developed the approach of tone enhancement that turned English speech recognizer to tonal language speech recognizer. Test was performed to compare recognition performance of both the speech recognizer in English and tone based on small database of Chinese names. Wong and Siu [133] presented the combination of pitch contour robust polynomial segmental representation with decision tree for tone modeling of Chinese SR as well as integrated decision tree tone classifier into single pass recognition process. 89% tone classification accuracy was for isolated syllable

and 71.2% tone classification accuracy was for continuous speech by the decision tree based classification. Zhang and Hirose [141] presented algorithm for lexical tone recognition in continuous Chinese speech. It was concluded that the CI tone nucleus approach was better than the CD approach in terms of recognition rate. Ma et al. [77] described the method of extracting frame based multi dimensional tone feature on the basis of pitch flux in Chinese continuous speech signal. The pitch flux features are combined with Mel-Frequency Cepstral Coefficient (MFCC) and modeled with Gaussian Mixture Model (GMM). 30% error rate was reduced when MFCC and pitch flux were fused as compared when only MFCC was used. Liu et al. [73] done the discussion and evaluation regarding the State-Dependent Phoneme Based Model Merging (SDPBMM) used for building Chinese speech recognizer from dialectal Chinese speech. Distance based pronunciation modeling approach was proposed to overcome the sparseness problem raised due to limited data at the state level. The experiments resulted that SDPBMM outperformed the Maximum Likelihood Linear Adaption (MLLR) in terms of Syllable Error Rate (SER). Wang and Levow [132] discussed the approach used to improve tone recognition of Chinese. Three baseline models called the whole, half and 60 ms were compared with the vowel landmark detection approach. The cl\_35 and cl\_10 vowel landmark method obtained mean of 65.31% and 65.31%, respectively. The baseline method whole, half and 60 ms got mean of 60.18%, 60.7% and 52.41%, respectively. Li et al. [67] for interactive Chinese SR proposed candidate generation method in which competitive candidates were in form of Chinese characters rather than words. The time-over linked links in Chinese word lattice was aligned into the clusters on the basis of the phonetic similarity and then the candidates were obtained by splitting the words that are attached to links. 94.06% character correct rate was obtained for all the top candidates. Yang et al. [137] used the combination of HMM and Conditional Random Field (CRF) for automatically generation of Chinese abbreviations as well as expanded the vocabulary expression for the voice search with the help of abbreviation model. 88.3% accuracy was achieved for abbreviation modeling and for full name 79.2% accuracy was obtained. Li and Wu [68] evaluated deep Long Short-Term Memory (LSTM) on Chinese large telephone speech recognition activity. The Character Error Rate (CER) of 34.65% was obtained by Long Short-Term Memory with Output Projection (LSTM-OP) layer. Li et al. [65] built Chinese speech recognition system with Long Short-Term Memory Projected (LSTMP) network architecture and Connectionist Temporal Classification (CTC). The LSTMP-CTC was compared with Deep Neural Network (DNN)-HMM on the basis of CER. The CER obtained by LSTMP-CTC was 13.9% and the CER was 12.6% for DNN-HMM. Li et al. [69] evaluated the performance of various units of acoustic

modeling with DNN for Chinese Large Vocabulary Continuous Speech Recognition (LVCSR) and compared it with GMM. The best CER observed by CD phone for GMM and DNN was 21.75% and 14.92%, respectively. Chen et al. [9] carried the analysis on phonetic and prosodic pronunciation pattern for evaluating the mispronunciations of Mandarin Chinese by non-native speakers of European origin and addressed the lack of large scale, non-native corpora with comprehensive annotations for an application in Computer Assisted Pronunciation (CAP) training. Zhou et al. [144] compared syllable-based model as well as CI phoneme model. The CER achieved by CI-phoneme model and syllable model was 30.72% and 28.77%, respectively. Zhou et al. [145] have done the comparison of CI phoneme, syllables, words, sub-words and characters with help of transformer on Chinese ASR activities. The CER for CI phoneme was found to be 30.65%.

## 5.2 Mandarin

Shen et al. [113] developed Mandarin syllable recognition algorithm for the selection of minimum set of phonetically distributed training sentences from a text corpus. The proposed approach was experimented on isolated syllable recognition. With speaker dependent 92.95% recognition accuracy was observed, and it was 94.14% for manually selected training syllables. Li [66] presented SR low cost computation method in which 408 Mandarin Monosyllables were classified and represented with sequence of Vectors of Linear Predict Coding Cepstra (LPCC). Reduction of computation for the classification was done with the help of Bayes Decision Rule with weighted variance. Lei et al. [64] proposed embedded approach and explicit tone model for improving tone modeling for Mandarin speech recognition based on broadcast news. 13.0% to 11.5% CER was obtained by combination of tone modeling approach on CTV test set. Chiang et al. [10] developed a model of Mandarin speech called the Latent Prosody Model (LPM) for automatic labeling the prosody state and for the enhancement of accuracy of tone recognition. Multi-Layer Perceptron (MLP) obtained 80.86% recognition rate and LPM model accounted 82.55%. Hoffmeister et al. [29] discussed the development of RWTH 2007 LVCSR system of Mandarin, in which various acoustic models were used as front ends 17.7% CER was obtained on evaluation data of GALE 2006. Ng et al. [81] reported the BBN 2007 progress for Mandarin Speech-To-Text (STT) system. CER by 25% relative reduction was obtained by BBN 2007 as compared to BBN 2006 Mandarin STT system. Hwang et al. [34] build Mandarin speech recognizer with help of language independent technologies as well as language dependent module and used MLP. CER of 9.1% was obtained on DARPA GALE 2007 project. Plahl et al. [94] developed LVCSR for Mandarin

within the project called GALE 2008 in which many types of Neural Network (NN) toneme posterior feature as well as decoding of two stage was used. The CER was 10.0% for GALE 2007 and 9.9% for GALE 2008. Qian and Soong [97] evaluated Multi-Space Distribution (MSD)-HMM and a two steam modeling approach on spontaneous Mandarin speech database. MSD-HMM as well as two steam modeling approach gained significant performance enhancement for the recognition of tonal syllables in Mandarin. Valente et al. [125] done the comparative study for Mandarin ASR on the MLP features. The CER obtained for GALE 2007 development corpus and GALE 2008 development corpus was 8.8% and 8.5%, respectively. Yu [138] analyzed laryngealization useful for the tone as cue in tonal perceptions. The laryngealization instances of tones in Cantonese and Mandarin corpora suggested that laryngealization was spread over domain of feature extraction for a syllable and concluded that laryngealization features should be included in tonal recognition. Amodei et al. [3] recognized the English and Chinese speech with the help of end-to-end deep learning approach and look ahead convolution approaches. 9.39% and 6.22% accuracy was obtained by Recurrent Neural Network (RNN) for noisy and clean ENVIRONMENT, respectively. Chan and Lane [7] evaluated online speech recognition using attention-based model on English as well as on Mandarin. The Word Error Rate (WER) obtained for English was 33.0% and Mandarin character model got a CER of 59.3%. Huang et al. [31] proposed encoder classifier framework in which RNN was used for modeling the tones in Mandarin and the comparison of RNN-DNN with GMM-HMM indicated that RNN-DNN gave improved tone classification accuracy. Shan et al. [112] discussed Mandarin SR on the basis of voice search activity with the help of attention based encoder-decoder model. The CER and Sentence Error Rate (SER) of 3.58% and 7.43% was obtained, respectively, on MiTV voice search. Zou et al. [146] for Mandarin SR explored CTC model as well as attention-based encoder decoder model. The CER achieved by CTC and attention-based model on DidiCallcenter dataset was 7.45% and 5.68%, respectively.

## 5.3 Thai

Ahkuputra et al. [2] reviewed the various SR on techniques like NN, Fuzzy-NN, HMM model and Modified Backpropagation NN. The recognition error rate for Fuzzy-NN was 8.50% and for NN it was 26.97. The HMM and Modified Backpropagation accounted recognition error rate of 15.75% and 22.0%, respectively. Jitapunkul et al. [35] addressed the work done on speech analysis as well as on SR of Thai. Potisuk et al. [95] presented an algorithm called Analysis by Synthesis for the classification of Thai tones in the syllable segmented speech in which the Fujjaski's model was used. The algorithm identified correctly all the tones in 89.1% of

test utterance. Thathong et al. [121] proposed a technique that used Thai tones to separate Thai consonant spelling as well as consonant naming. It was concluded that in the case of female voices the  $f_0$  was increased in rising tone as well as in mid tone, but it remains constant in the case of male voice. Thubthong and Kijirikul [123] proposed a model called Half-Tone model (H-T-30) for Thai continuous tone recognition. The comparison of tone models CI (CI-T-5) and CD (CD-T-175) was done. H-T-30 outperformed better than CI and CD in term of recognition rate and speed. Karnjanadecha and Kimsawad [43] done the comparison on the performance of Linear Predictive Cepstral Coefficient (LPCC), MFCC and Discrete Cosine Transform Coefficient (DCTC) with the aim to develop a front end analysis for Thai SR. Maneenoi et al. [78] proposed new speech unit called onset rhyme unit for Thai SR. The corpus contained 57 phoneme units, 275 onsets and 159 rhymes. The SER was 32.91% for onset-rhyme. Wutiwiwatchai et al. [134] focused on the design of Thai speech corpus based on Phonetically Distributed (PD) sentences set and Phonetically Balanced (PB) sentences as well as on the procedure of selection of PD sentences set and PB sentences. Thai PB sentences set covered 90.9% biphones as compared to Japanese ATR's 503 which covered 87.5%. Kasuriya et al. [46] discussed the design of the Thai speech corpus as well as the corpus tool and the problem of word variation in Thai language. The corpus contained the PD sentences set and 5000 vocabulary coverage sentences set. Automatic segmentation, automatic sentences distributor and wave cutting tool was used as corpus tool. Kasuriya et al. [44] did the investigation as well as construction of various acoustic models which included acoustic modeling with tone and acoustic model without tone for Thai speech. The Word Accuracy (WA) without tone model, the single phone model and base phone model obtained was 87.03% and 83.13%, respectively. Single phone model and base phone model with tone model obtained the WA of 78.28% and 82.03%, respectively. In the same year Kasuriya et al. [45] discussed the design of Thai speech corpus ORCHID and NECTEC-ATR. The ORCHID corpus contained the 5000 Thai sentences and PD sentence set. Pisarn and Theeramunkong [90] constructed Thai continuous speech recognition system by merging the spelling recognition system with the conventional recognizer. In the training module the speech Utterances of the Corpus were transformed to the feature vectors. Pisarn et al. [93] constructed the spelling recognition system with the help of continuous speech corpus for Thai using HMM. The performance of the spelling method was investigated under the open, mix and closed type ENVIRONMENT. Suebvisai et al. [119] discussed the Thai speech recognizer development using limited resources related to tone and data which being the part of DARPA Babylon project. WER for Babylon and Global Phone (GP) with tone modeling was 18.5%

and 16.2% respectively. Pisarn and Theeramunkong [92] proposed a method based on HMM for Thai spelling SR with the use of Thai alphabets sets and the proposed system performance was evaluated with bigram and trigram in the environment of close type, open type and mix type. [135] reviewed all the work done on Thai Speech as well on the Thai language sound system. Bhuriyakorn et al. [6] proposed an estimation approach for Thai phoneme recognition based on HMM topology. Genetic Algorithm (GA) was used as the selection algorithm for the selection of the most suitable topology. Jongtaveesataporn et al. [37] presented the construction as well as evaluation of news speech of 17 h and text corpora of 35 h for Thai for first time. Baseline system without adaptation obtained WER of 36.5% for male speakers and it was 41% for female speakers. The WER when focus condition adapted with the acoustic model was 30.9% and 32.6% for male and female speakers, respectively. Chotimongkol et al. [11] described the design and the construction of corpus called LOTUS-Broadcast News (BN) which was related to Thai Television BN. It contained the BN speech of 100 h. The percentage of all the utterance that contained the overlapping speech was 3.7%. Jongtaveesataporn et al. [38] merged Pseudo-Morpheme (PM) with Compound Pseudo-Morpheme (CPM) for solving the problem of Thai words ambiguous definition. Predawan et al. [96] presented the method that recognized the monosyllabic Thai tone with the help of Anti Miner algorithm. The recognition rate achieved by Anti Miner algorithm was 96% and Karhunen–Loeve Transformation (KLT)-Learning Vector Quantization (LVQ) obtained 90% of recognition rate. Sodanil et al. [116] cited the approach for enhancing the ASR of Thai continuous words based on Artificial Neural Network (ANN). The acoustic model was trained with MFCC as well as with tone information as input vectors. The comparison between ANN and GMM was done. ANN yielded higher accuracy and lower WER as compared to GMM. Theera-Umpon et al. [122] recognized Thai phonemes that included 21 initial consonants, 18 vowels and 9 final consonants using HMM, MFCC and Mel Frequency with Perceptual Linear Prediction (MFPLP). The tonal accents of Thai were recognized with the help of NN. The MFPLP gave better result as compared to MFCC. Kertkeidkachorn et al. [49] discussed the spelling speech recognition corpus called CU-MFEC for Thai and English. Spelling task obtained accuracy of 79.37% and accuracy achieved by connected task was 54.92%. Srisuwan et al. [118] developed Thai tone SR system based on Electromyography (EMG). Top five features out of 68 features on the basis of RES index were selected and NN obtained 80% accuracy for Thai tones. Hu et al. [30] explored Mandarin and Thai ASR system by including tone features into Convolutional Neural Network (CNN) and compared CNN with DNN. The WER for Thai was 33.19% and 35.16% for CNN and DNN, respectively,

for the VoiceTra corpus. Kertkeidkachorn et al. [50] discussed the spelling SR system for Thai in which the SR system's first pass search N-best hypothesis was re-ranked by the tonal score. Hidden Conditional Random Field (HCRF) represented tonal score and accounted 87.93% accuracy whereas HMM obtained 84.15% of accuracy. Tantibundhit and Onsuwan [120] done the designing as well as development of Subjective Intelligibility Testing for Thai Speech for initial consonants called Thai Diagnostic Rhyme Test for Initials (TDRT-I) based on Diagnostic Rhyme Test (DRT). Perception of all phoneme was improved by 10%. Srijiranon and Eiamkanitchat [117] presented a comparison of Neuro-Fuzzy algorithm with Decision Tree, Bayes, GMM and Support Vector Machine (SVM) on Thai words in continuous and easy control noise and loud and unpredictable noise. The Neuro-Fuzzy achieved higher accuracy in continuous and easy control noise as compared to Decision Tree, Bayes, GMM and SVM. But in loud and unpredictable noise, SVM achieved higher accuracy as compared to Decision Tree, Bayes, GMM and Neuro-Fuzzy.

## 5.4 Vietnamese

Dat et al. [13] presented the method for calculating the influence of f0 on Vietnamese syllable perception. Synthesis of arbitrary syllables was done and the perception tests which helped in understanding as well as defining the role played by f0 on Vietnamese tones characterization were conducted. Le et al. [63] gave the description about the steps carried out in building LVCSR system for Vietnamese. Phoneme Correct Rate (PCR) and WA was calculated for Vietnam test data of 1 h collected from 25 females and 25 males' speakers. Le and Besacier [60] described the acoustic modeling for Vietnamese where the Vietnamese was treated as target language and French as the source language. The data driven method slightly gave better results than the knowledge based method without the adaptation data in the model mapping approach. 63.9% WA was obtained for Vietnamese. Besacier et al. [5] presented an overview of the steps concerned with ASR as well as speech-to-speech translation for Vietnamese and Khmer language. Word accuracy of 73.6% was obtained for Khmer ASR. Syllable Accuracy (SA) of 57% was obtained for Vietnamese speech. Dat et al. [14] used the perception test and TD-PSOLA algorithm for the speech synthesis of Vietnamese. The effect of f0 on Vietnamese tone. The perception test conducted on 4 groups having 84 different syllables. Natural group syllable recognize tone by correct rate of 99.05% and recognize syllable name by 96.79%. Le and Besacier [61] cited the comparison of Vietnamese ASR and Khmer ASR. 8% of segmented word error and 4.0% of segmented sentence error obtained for Vietnamese and Khmer language, respectively. Vu et al. [127] worked for Vietnamese ASR at the word level with

new method based on FLAVOR approach, the first layer contained phoneme network with meta data and the second layer was concerned with the word decoding and combining of various sub-lexical, word lexical and word based Language Model (LM). Quang et al. [99] analyzed the usage of tone information as well as tone recognition of Vietnamese continuous speech. VNSPEECHCORPUS was used. The tone recognition model was developed with the help of HMM and integrated into the system called Speeral. The average Tone Accuracy Rate (TAR) was 75.08%. In the same year Quang et al. [100] presented a study for constructing Vietnamese LVCSR. The average Tone Correct Rate (TCR) for all the six tones was 44.8%. Le and Besacier [62] worked for Vietnamese ASR and investigated the proposed grapheme-based acoustic model in which building of phonetic dictionary was avoided. Word/Sub word lattices decomposition and combination approach was proposed. 6.6% reduction of Syllable Error Rate (SLER) was observed by lattice combination method. Vu et al. [129] developed corpus for Vietnamese which contained the read speech, Basic Travel Expression Corpus (BTEC), speech synthesis and broadcast corpus. The WER of SR was 13.85% and BLEU score of Machine Translation (MT) was 0.5826. Vu and Schultz [128] used rapid language adaption schemes for bootstrapping the Vietnamese SR. 25 h speech data was recorded. 12.6% was the improved WER on the development set and 11.7% on the evaluation set. Quang et al. [98] build Vietnamese SR system using HMM. The data for the corpus was collected from the web. Nguyen and Tran [83] conducted a study to find out the influences of tonal co-articulation of particles in Vietnamese. The study resulted that tonal co-articulation in Vietnamese always occur at transition between the two adjacent syllables. Chuong and Chaloupka [12] developed the text and speech data corpus for Vietnamese ASR. The data was collected from the websites related to news and literature of Vietnamese. It involved the collection of the small text corpus followed by selection of the seed words from the text corpus and then these seeds were fed into the next stage for collecting the large and diverse text corpus. Chen et al. [8] proposed Keyword Search (KWS) for Vietnamese from an open source tool at Institute for Infocomm Research (I2R) which find out acoustic features having creaky voice related to lexical tone. The Actual Term Weighted Value (ATWV) was 0.47. Huy et al. [33] presented MSD-HMM for recognition of Vietnamese on the basis of Vietnamese tonal phonemes. The MSD-HMM gave 80.80% of WA and HMM got WA of 78.31%. Karafiat et al. [42] presented Vietnamese SR that contained spontaneous telephone conversations. The Stacked Bottle Neck (SBN), NN were used. Hung et al. [32] build VDSPEC Vietnamese speech corpus which served the purpose of identifying Vietnamese dialects and Vietnamese synthesis. Nguyen et al. [84] proposed algorithm that did the transformation of any Vietnamese word into pronunciation

of tonal phoneme. The WER for VoiceTra test was 27.73% and for BTEC test, it was 9.14%. Nguyen et al. [82] used Deep Bottle Neck Features (DBNFs) in Vietnamese LVCSR system as well as used the tonal features. The DBNFs gave 10.76% of WER and WER achieved by HMM-GMM was 15.56%.

### 5.5 Mizo

Govind et al. [24] studied the pitch slope role and duration in synthesizing Mizo tones and obtained the correctness score of 84%, 63% and 98% for recognizing real speech for falling tone, low tone and rising tone, respectively. Sarma et al. [109] proposed automatically method to detect level tone or contour tone in Mizo. Classification was done based on pitch values as well as based on height parameters obtained from pitch values fitted to tone. Dey et al. [15] discussed the building of speech corpus and phone recognition system for Mizo. DNN and Subspace(S)GMM were used for the development of Mizo phone recognition system. The Phone Error Rate (PER) for DNN was 23.2% and it was 10.3% for SGMM (Tables 4, 5, 6).

## 6 Reported Work on African Tonal Languages

### 6.1 Yoruba

Odelobi [86] addressed the ANN approach for tone recognition of Yoruba. RNN achieved recognition rate of 76% and obtained recognition rate of 71% for high tone. Yusof et al. [139] reviewed the work done on Yoruba ASR and also studied the performance gap between the Human SR and the ASR for Yoruba. Fagbolu et al. [19] discussed the digital corpus of Yoruba which was built with three modules. The text processing module take the Yoruba words as input and converted them into sequence of phonetic description of high level. PSOLA algorithm was applied to prosodic features. Adetunmbi et al. [1] presented a syllable-based recognition system for Yoruba. The 25 bi-syllabic, 25 tri syllabic words were recorded using PRAAT software.

### 6.2 Hausa

Luka et al. [76] presented ANN based Hausa SR. The neural network was trained with the MFCC. The database size was of 320 words, and the words were uttered by the two male and two female speakers. The performance of speaker 2nd and speaker 4th were observed better than speaker 1st and 3rd. Better generalization was achieved with combination of individual speakers. Schlippe et al. [110] discussed Hausa speech recognition system for large vocabulary with

the help of Rapid Language Adaptation Toolkit (RLAT). The sources of the corpus was from many websites and GP style was followed.

## 7 Reported Work on Indo-European Tonal Languages

### 7.1 Croatian

Vu et al. [126] used the RLAT for developing the LVCSR for Bulgarian, Croatian, Czech, Polish and Russian. Gulic et al. [27] developed the Croatian speech corpus that contained alphanumeric utterances and also built the digit and spelling SR system for Croatian. Seljan and Dunder [111] presented the combination for ASR of English as well as MT for English and Croatian and Croatian to English for business purpose. For English the WER was 49.00%. For Croatian WER was 67.89%. Martincic-Ipsic et al. [79] developed the bilingual spoken dialog system in Slovenian and Croatian for weather forecast. Combination of both languages into one bilingual SR system does not resulted loss in accuracy. The 91.33% accuracy was obtained when ASR of both languages was merged. Nouza et al. [85] did the study of porting ASR of Czech into Croatian. WER was 27.6% for Czech and it was noted 19.4% when it was ported to Croatian. Ljubetic et al. [74] discussed the Croatian reference training corpus hr500k, that contained 5000 thousand tokens, sentences, word level dependency syntax, named entities, semantic roles.

### 7.2 Swedish

Ohman [87] recorded the audio-visual Swedish speech database and the jaw movements were easily traced with constructed device. For lip's shape and speaker's face visual feature algorithm was developed. Hallgren et al. [28] cited the usage of Hearing in Noise Test (HINT) for estimating the threshold of SR in noise and quiet. In Noise the average threshold for Swedish sentences as signal/noise ratio was  $-3.0$  dB. Lu et al. [75] performed the experiment on SGMM using GP corpus for SR for cross lingual modeling where the target language was German has limited training data and the source language were Spanish, Portuguese, Swedish. Lindh and Eriksson [72] discussed the SweDia 2000 project that served the purpose of transforming database into e-science database. The database contained Swedish dialect recording collected from 107 location of Sweden.

### 7.3 Lithuanian

Raskinis and Raskiniene [103] described the process of developing the HMM based SR system for Lithuanian. The

**Table 4** Some of important work on Asian tonal languages

| Author                          | Year | Language   | Dataset                                                                                                                            | Feature extraction                                         | Classifier           | LM/ENV.                                                                     | Result                                                                                                     |  |  |
|---------------------------------|------|------------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|----------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|--|--|
| Wang [130]                      | 2000 | Chinese    | BCC Corpus<br>Speaker: 20<br>Documents: 757<br>Duration: 10.2 h                                                                    | MFCC                                                       | HMM                  | VSM<br>SLM<br>VSM, SLM                                                      | Average Precision<br>TQ/TD TQ/SD SQ/TD<br>0.917 0.5213 0.4228<br>– 0.5998 0.5278<br>– 0.6007 0.5255        |  |  |
| Thubthong and Kijisirikul [123] | 2001 | Thai       | Utterance: 550<br>Speakers: 5 males, 5 females                                                                                     | A, D                                                       | NN                   | CI-T-175<br>CD-T-5<br>H-T-30                                                | Accuracy (%)<br>Inside test Outside test<br>94.77 93.00<br>93.27 92.09<br>94.77 93.82                      |  |  |
| Karnjanadecha and Kimsawad [43] | 2002 | Thai       | Utterances: 5 different 9 digit strings uttered in PS1 style and PS2 style<br>Speakers: 40 males, 40 females                       | DCTC<br>MFCC                                               | HMM                  | –                                                                           | Accuracy (%)<br>94.8<br>99.6                                                                               |  |  |
| Tsai and Chang [124]            | 2002 | Chinese    | Speaker: 5 males, 3 females<br>Training<br>Utterances: 480<br>Duration: 7200 s<br>Testing<br>Utterances: 240<br>Duration: 3600 s   | Mel-cepstral<br>Pitch-based<br>Mel-cepstral<br>Pitch-based | GMM                  | ML<br>MCE                                                                   | Recognition (%)<br>Mandarin Holo Hakka<br>83 88 81<br>54 58 59<br>91 90 87<br>56 62 62                     |  |  |
| Zhang et al. [143]              | 2002 | Chinese    | Surnames<br>Single character: 2626<br>Double character: 55                                                                         | MFCC                                                       | HMM                  | –                                                                           | Accuracy (%)<br>95.97 (Surnames)<br>95.88 (Single character surnames)<br>96.15 (Double character surnames) |  |  |
| Li [66]                         | 2003 | Mandarin   | Syllables: 408                                                                                                                     | LPCC                                                       | Decision Tree<br>HMM | –                                                                           | Recognition Rate (%)<br>94.13<br>79.78                                                                     |  |  |
| Pisarn and Theeramunkong [90]   | 2004 | Thai       | NECTEC-ATR corpus<br>Utterances: 3900<br>Speakers: 21 males, 21 females                                                            | PLP                                                        | HMM                  |                                                                             | Accuracy (%)<br>87.18<br>Correctness (%)<br>87.37                                                          |  |  |
| Zhang and Hirose [141]          | 2004 | Chinese    | HKU96 corpus<br>Speaker: 1 female<br>Training<br>Utterances: 500<br>Testing<br>Utterances: 200                                     | LDA                                                        | HMM                  | Testing<br>CI nucleus<br>CD nucleus<br>Training<br>CI nucleus<br>CD nucleus | Recognition rate (%)<br>80.8<br>91.8<br>81.5<br>83.1                                                       |  |  |
| Le et al. [63]                  | 2005 | Vietnamese | Speakers: 15<br>Duration: 3 h<br>Phonemes, tones, digits, applications words, short dialogues                                      | MFCC                                                       | Gaussian mixture     | 1st Bootstrapping Cycle<br>2nd Bootstrapping Cycle                          | PCR (%) WA (%)<br>59.37 55.3<br>62.87 55.5                                                                 |  |  |
| Pisarn et al. [93]              | 2005 | Thai       | Speakers: 21 males, 21 females.<br>LM1: 136 proper names<br>LM2: 136 proper names, 5971 location names<br>LM3: 5971 location names | PLP                                                        | HMM                  | Tri-gram model<br>LM1<br>LM2<br>LM3                                         | COR (%) ACC (%)<br>93.19 93.08<br>84.65 83.92<br>91.12 90.80                                               |  |  |
| Pisarn and Theeramunkong [91]   | 2006 | Thai       | Speakers: 21 males, 21 females.<br>LM1: 136 proper names<br>LM2: 136 proper names, 5971 location names<br>LM3: 5971 location names | PLP                                                        | HMM                  | Tri-gram model<br>LM1<br>LM2<br>LM3                                         | COR (%) ACC (%)<br>80.08 77.01<br>73.04 68.17<br>71.55 66.36                                               |  |  |
| Vu et al. [127]                 | 2006 | Vietnamese | VNBN corpus<br>Sentences: 19,496<br>Duration: 20 h                                                                                 | GMM                                                        | MFCC                 | Baseline<br>Flavor method                                                   | WU error rate<br>19.1<br>17.8                                                                              |  |  |
| Pisarn and Theeramunkong [92]   | 2007 | Thai       | SPELL-1<br>Speakers: 3 males and 3 females<br>Utterances: 900<br>SPELL-2<br>Speakers: 5 males and 5 females<br>Utterances: 3900    | PLP                                                        | HMM                  |                                                                             | LCR (%)<br>93.69<br>81.12                                                                                  |  |  |

**Table 4** (continued)

| Author                         | Year  | Language   | Dataset                                                                                                             | Feature extraction              | Classifier               | LM/ENV.                                                                              | Result                                                                                                                        |
|--------------------------------|-------|------------|---------------------------------------------------------------------------------------------------------------------|---------------------------------|--------------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Bhuriyakorn et al. [6]         | 2008  | Thai       | LOTUS-BN corpus<br>Sentences: 500<br>Speakers: 124 males, 124 females                                               | MFCC                            | GA                       | –                                                                                    | Accuracy (%)<br>63.66                                                                                                         |
| Liu et al. [73]                | 2008  | Chinese    | MBN Corpus<br>Utterances: 1200<br>Duration: 80 min<br>WDC corpus<br>Speaker: 20 Utterances: 995<br>Duration: 60 min | MLLR<br>MLLR + SDPBMM           | HMM                      | –                                                                                    | SER (%)<br>MBN WDC<br>43.8 44.1<br>42.2 41.3                                                                                  |
| Quang et al. [101]             | 2008c | Vietnamese | VNSpeech Corpus<br>Speakers: 10 men, 8 women<br>Duration: 14.4                                                      | –                               | HMM                      | Norm_Log_F0_<br>Mean_Dev<br>Men and Women<br>8 tone forms                            | TAR (%)<br>70.44<br>72.83<br>81.02                                                                                            |
| Quang et al. [102]             | 2008d | Vietnamese | VNSpeech Corpus<br>Speakers: 10 men, 8 women<br>Duration: 14.4                                                      | –                               | HMM                      | LM_DP_MI<br>with tones<br>LM_MM with tones                                           | SER (%)<br>18.7 (Speaker Independent)<br>10.7 (Speaker adaptation)<br>21.2 (Speaker Independent)<br>12.4 (Speaker adaptation) |
| Qian and Soong [97]            | 2009  | Mandarin   | BJ2003 Corpus<br>Speaker: 490<br>Utterances: 4000                                                                   | MFCC<br>MSD<br>MFCC<br>MSD      | HMM                      | Bi-gram model<br>Tri-gram                                                            | CER (%)<br>Male Female<br>16.91 12.85<br>14.03 10.56<br>12.09 8.71<br>10.55 7.21                                              |
| Jongtaveesa-taporn et al. [38] | 2009  | Thai       | Thai Broadcast News<br>Speech Corpus<br>Utterances: 1033<br>Duration: 40.3 h<br>Speakers: 68 males, 68 females      | MFCC                            | HMM                      | PM<br>CPM                                                                            | PER (%) OOV (%)<br>23.7 0.2<br>22.1 0.4                                                                                       |
| Quang et al. [98]              | 2010  | Vietnamese | Utterances: 8047<br>Speakers: 23<br>Duration: 93 h                                                                  | MFCC<br>MLLR                    | HMM                      | –                                                                                    | WA (%)<br>71.37 (Independent Speaker)<br>75.96 (Dependent Speaker)                                                            |
| Sodanil et al. [116]           | 2010  | Thai       | Words: 809<br>Syllables: 2787<br>Speakers: 10                                                                       | MFCC + TF                       | GMM<br>ANN               | 3-gram                                                                               | WER (%)<br>24.76<br>23.35                                                                                                     |
| Theera-Umpon et al. [122]      | 2011  | Thai       | Sentences: 3045<br>Speakers: 15<br>Duration: 88 h 8 min.                                                            | MFCC<br>MFPLP<br>MFPLP<br>Pitch | HMM<br>NN                | Speaker-Dependent<br>Speaker-Independent<br>Speaker-Dependent<br>Speaker-Independent | Classification rate (%)<br>Vowels<br>67.71<br>67.86<br>67.82<br>Tonal accents<br>89.69<br>89.14                               |
| Govind et al. [24]             | 2012  | Mizo       | Speakers: 1<br>Duration: 40 min                                                                                     | –                               | –                        | ENV.<br>Real                                                                         | Correctness (%)<br>Falling tone: 84.6<br>Low tone: 63<br>Rising tone: 98                                                      |
| Kertkeidkachorn et al. [49]    | 2012  | Thai       | Speakers: 100<br>Words: 16,748<br>Duration: 58 h 10 min                                                             | MFCC                            | HMM                      | –                                                                                    | Accuracy (%)<br>Connected alphabets: 79.39<br>Continuous spelling: 54.92                                                      |
| Yang et al. [137]              | 2012  | Chinese    | Speakers: 20<br>Abbreviations: 2200<br>Organizationfullname: 1945<br>Duration: 150 h                                | MFCC                            | HMM-CRF                  | ENV.<br>Clean                                                                        | Accuracy (%)<br>Abbreviation modeling: 88.3<br>Fullname: 79.2                                                                 |
| Hu et al. [30]                 | 2014  | Thai       | Mandarin<br>HUB4:<br>Read speech<br>Duration: 27 h.<br>Thai: VoiceTRA:<br>Network speech<br>Duration: 100 h         | MFCC                            | CNN<br>DNN<br>CNN<br>DNN | –                                                                                    | CER (%)<br>10.92<br>12.57<br>WER (%)<br>33.19<br>35.16                                                                        |

**Table 4** (continued)

| Author                             | Year  | Language   | Dataset                                                                                                                                 | Feature extraction                        | Classifier                                                                                                       | LM/ENV.                                                                       | Result                                                                                                               |  |  |  |
|------------------------------------|-------|------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|--|--|--|
| Huy et al. [33]                    | 2014  | Vietnamese | VOV<br>Speakers: 30<br>Words: 23,424,<br>Distinct syllables with<br>tone: 4923<br>Duration: 19 h                                        | MFCC<br>MFCC + NCC + AMDF_<br>MSD         | HMM<br>MSD-HMM                                                                                                   | –                                                                             | WA (%)<br>78.31<br>80.80                                                                                             |  |  |  |
| Karafiat et al. [42]               | 2014  | Vietnamese | Scripted telephone speech<br>Conversational spontaneous<br>telephone conversations                                                      | MFCC                                      | NN<br>DNN                                                                                                        |                                                                               | WER (%)<br>48.9<br>45.5                                                                                              |  |  |  |
| Li and Wu [68]                     | 2015  | Chinese    | HKUST Corpus<br>Mandarin<br>Chinese telephonic conversation<br>Duration: 150 h                                                          | Mel-Scale log filter<br>bank coefficients | DNN<br>LSTM-OP                                                                                                   | –                                                                             | CER (%)<br>38.42<br>34.65                                                                                            |  |  |  |
| Li et al. [69]                     | 2015b | Chinese    | Hub-4NE corpus<br>Broadcast news of Chinese<br>Duration: 30 h                                                                           | MFCC                                      | GMM<br>DNN                                                                                                       | CD Model<br>Phones                                                            | CER (%)<br>21.75<br>14.92                                                                                            |  |  |  |
| Nguyen et al. [84]                 | 2015  | Vietnamese | VoiceTra test<br>Words: 803<br>Duration: 36 min<br>BTEC test<br>Words: 510<br>Duration: 19 min                                          | BNF                                       | GMM-<br>HMM                                                                                                      | –                                                                             | WER (%)<br>27.73 (VoiceTra test)<br>9.14 (BTEC test)                                                                 |  |  |  |
| Sarma et al. [109]                 | 2015  | Mizo       | Speakers: 5<br>Syllable: 4384<br>which contained<br>672 high tones<br>2254 low tones<br>808 falling tones<br>650 rising tones           | –                                         | –                                                                                                                | ENV<br>Real                                                                   | DR (%)<br>Without CE    with CE<br>50.14    35.26<br>81.28    56.78<br>75.23    78.92<br>52.47    74.62              |  |  |  |
| Srijiranon and Eiamkanitchat [117] | 2015  | Thai       | Speakers: 7 males and 7<br>females<br>Words: 9                                                                                          | PLP                                       | Decision<br>Tree<br>Bayes<br>GMM<br>SVM<br>Neuro-Fuzzy<br>Decision<br>Tree<br>Bayes<br>GMM<br>SVM<br>Neuro-Fuzzy | Continuous and<br>Easy control<br>Noise<br>Loud and<br>Unpredictable<br>noise | Accuracy (%)<br>53.70<br>56.13<br>73.76<br>76.57<br>84.85<br>36.57<br>30.06<br>52.66<br>60.50<br>59.81               |  |  |  |
| Amodei et al. [3]                  | 2016  | Mandarin   | Mandarin<br>Labeled speech<br>Words: 11 M                                                                                               | End-to-end RNN                            | RNN                                                                                                              | Noisy<br>Clean                                                                | Accuracy (%)<br>9.39<br>6.22                                                                                         |  |  |  |
| Chan and Lane [7]                  | 2016  | Mandarin   | For English<br>Wall Street Journal<br>For Mandarin<br>GALE<br>104 h speech                                                              | End to end RNN                            | RNN                                                                                                              | English<br>Online CTC<br>Online<br>CTC + Pinyin                               | WER (%)<br>33.0<br>CER (%)<br>59.3                                                                                   |  |  |  |
| Chen et al. [9]                    | 2016  | Mandarin   | iCALL corpus-<br>Speakers: 305<br>Words: 90,841<br>Duration: 142 h<br>English speakers<br>French Speakers                               | –                                         | –                                                                                                                | –                                                                             | Lexical tone accuracy (%)<br>Tones<br>1    2    3    4<br>73.0    71.4    62.4    72.2<br>71.2    57    53.5    57.8 |  |  |  |
| Nguyen et al. [82]                 | 2016  | Vietnamese | Speakers: 30<br>Words: 23,424<br>Duration: 19 h                                                                                         | MFCC                                      | HMM-<br>GMM<br>DBNFs                                                                                             | –                                                                             | WER (%)<br>15.56<br>10.76                                                                                            |  |  |  |
| Dey et al. [15]                    | 2017  | Mizo       | For training<br>Speakers: 62<br>Words: 7394<br>Duration: 5 h 5 min<br>For testing<br>Speakers: 19<br>Words: 1568<br>Duration: 1 h 5 min | MFCC                                      | DNN<br>SGMM                                                                                                      | –                                                                             | PER (%)<br>23.2<br>10.3                                                                                              |  |  |  |

**Table 4** (continued)

| Author            | Year  | Language | Dataset                                                                                                                                                        | Feature extraction    | Classifier                | LM/ENV.                                                     | Result                           |         |
|-------------------|-------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------------|-------------------------------------------------------------|----------------------------------|---------|
| Huang et al. [31] | 2017  | Mandarin | Speakers: 160<br>Words: 92,243<br>Duration: 110 h                                                                                                              | LDA-MLLT              | RNN<br>DNN-RNN<br>GMM-HMM | –<br>MLE<br>MMI                                             | Tone classification accuracy (%) |         |
|                   |       |          |                                                                                                                                                                |                       |                           |                                                             | 70.7                             |         |
|                   |       |          |                                                                                                                                                                |                       |                           |                                                             | 70.4                             |         |
|                   |       |          |                                                                                                                                                                |                       |                           |                                                             | 75.7                             |         |
| Shan et al. [112] | 2018  | Mandarin | Words: 4 M approx.<br>Duration: 3000 h                                                                                                                         | Mel-scale filter bank | BLSTM                     | –                                                           | CER (%)                          | SER (%) |
|                   |       |          |                                                                                                                                                                |                       |                           |                                                             | 3.58                             | 7.43    |
| Zou et al. [146]  | 2018  | Mandarin | DidiCallcenter<br>Spontaneous data<br>Words: 2.2 M<br>Duration: About 2800 h<br>DidiReading<br>Reading style data<br>Words: 16.2 M<br>Duration: About 12,000 h | Mel-scale filter bank | BLSTM<br>LSTM             | Attention model<br>CTC<br>Attention model<br>CTC            | CER (%)                          |         |
|                   |       |          |                                                                                                                                                                |                       |                           |                                                             | 5.68                             |         |
|                   |       |          |                                                                                                                                                                |                       |                           |                                                             | 7.45                             |         |
|                   |       |          |                                                                                                                                                                |                       |                           |                                                             | 4.89                             |         |
|                   |       |          |                                                                                                                                                                |                       |                           |                                                             | 5.79                             |         |
| Zhou et al. [144] | 2018a | Chinese  | HKUST corpus<br>Duration: 150 h<br>Chinese<br>telephone speech                                                                                                 | Log-Mel filter bank   | –                         | CI-phoneme-<br>Syllable-                                    | CER (%)                          |         |
|                   |       |          |                                                                                                                                                                |                       |                           |                                                             | 30.65                            |         |
|                   |       |          |                                                                                                                                                                |                       |                           |                                                             | 28.77                            |         |
| Zhou et al. [145] | 2018b | Chinese  | HKUST corpus<br>Chinese<br>telephone speech<br>Duration: 150 h                                                                                                 | Log-Mel filter bank   |                           | CI phoneme<br>Syllables<br>Words<br>Sub words<br>Characters | CER (%)                          |         |
|                   |       |          |                                                                                                                                                                |                       |                           |                                                             | 30.65                            |         |
|                   |       |          |                                                                                                                                                                |                       |                           |                                                             | 28.77                            |         |
|                   |       |          |                                                                                                                                                                |                       |                           |                                                             | 27.42                            |         |
|                   |       |          |                                                                                                                                                                |                       |                           |                                                             | 27.26                            |         |
|                   |       |          |                                                                                                                                                                |                       |                           |                                                             | 26.64                            |         |

WER was 20%. Silingas et al. [114] presented the investigation on acoustic modeling for Lithuanian continuous SR on the basis of simple as well as contextual phone set.

Laurinciukaite et al. [58] discussed the design and development of the speech corpus Lithuanian Radio News Prototype Version 0.1 (LRN 0.1). 31 speakers read the BN and total of 17 h and 23 min data for corpus was recorded. The segmentation of recorded material into sentence level was done. Laurinciukaite et al. [57] developed continuous Lithuanian speech corpus, Lithuanian Radio News Version 1 (LRN1) of 20 h and 50 min and contained various level of annotation carried by two-stage process, methods as well as tools. Maskeliunas et al. [80] checked the accuracy of the SR of 10 Lithuanian digits with the help of speech recognizers of English, German, French and Spanish. The accuracy of 93.5%, 64.8%, 67.7% and 55.1% for the Lithuanian digits were achieved by the Spanish, French, English and German speech recognizers respectively. Rudzionis et al. [106] presented the Lithuanian voice commands for controlling the medical pharmaceutical information system and used the HMM classifier and Spanish speech recognizer and utilized the Ripper algorithm. The recognition achieved for Lithuanian medical pharmaceutical voice command was 98.9%. Rasytas and Rudzionis [105] done the combinations of various recognizers outputs to improve speech recognizer by combining Lithuanian speech recognizer with Russian English, German's speech recognizer and used the text transcribing method based on format rules. Lileikyte et al. [70] cited the development of Lithuanian STT transcription

system for broadcast data and done the investigation of unsupervised acoustic model training. The Quaero program was used and 18.3% WER was obtained on the development data set. Greibus et al. [25] presented the influence of phoneme set for determining the accuracy of Lithuanian speech commands recognition. Acoustic models were trained with LIEPA speech corpus and for Lithuanian language rule-based transformation was proposed. The phoneme set 36 (FZ13) produced error rate –3.76% and perform better than other phoneme sets. Lileikyte et al. [71] developed system for Lithuanian called Conversational Telephone Speech (CTS) as well as Keyword Spotting (KWS). The experiments were conducted on two conditions called Full Language Pack (FLP) that contained the transcribed speech of 40 h for the training purpose and the Very Limited Language Pack (VLLP) that was the subset of FLP. Laurinciukaite et al. [59] cited the problem of LIEPA, Lithuanian speech corpus related with designing of human–computer interface that work in voice recognition and in the synthesis mode. The first part of corpus contained requirements of Lithuanian text-speech synthesizers and second part contained the Lithuanian speech SR engines.

## 7.4 Latvian

Oparin et al. [88] discussed the development of STT system within Quaero project for Latvian For BN and for conversation data WER of 20% was obtained. Pinnis et al. [89] discussed the development of first speech corpus of 100 h

**Table 5** Important Work on African Tonal Languages

| Author                | Year | Language | Dataset                                                                           | Feature Extraction | Classifier | LM/ENV. | Result                                                                                                                     |
|-----------------------|------|----------|-----------------------------------------------------------------------------------|--------------------|------------|---------|----------------------------------------------------------------------------------------------------------------------------|
| Odelobi [86]          | 2008 | Yoruba   | Speakers: 3<br>Words: 350                                                         | MFCC               | MLP<br>RNN | –       | Tone Recognition Rate (%)<br>High tone   Mid tone   Low tone<br>71.30      85.50      81.20<br>76.10      86.15      83.42 |
| Schlippe et al. [110] | 2012 | Hausa    | Speakers: 33<br>male, 69<br>female<br>Utterances: 7895<br>Duration: 8 h<br>44 min | MFCC               | CNC        | RLAT    | WER (%)<br>13.6 (Development set)<br>16.26 (Test set)                                                                      |
| Adetunmbi et al. [1]  | 2016 | Yoruba   | Speakers: 6<br>Words:<br>Bi-syllabic: 75<br>Tri-syllabic: 50                      | MFCC               | HMM        | –       | Accuracy (%)<br>76 (Bi-syllabic)<br>84 (Tri-syllabic)                                                                      |

and 1 min developed for Latvian SR. The corpus contained the speech styles i.e. inhalation, exhalation, paused, non verbal, verbal segments. Salimbajevs and Strigins [108] performed detailed study on error analysis in Latvian with the help of Weighted Finite State Transducer (WFST). The WER achieved was 28.43%. Skrabal and Benko [115] cited the InterCorp (IC) parallel, large size corpus and Treq V2.0 database for Latvian. Safarik and Mateju [107] discussed Latvian ASR. acoustic model was developed with the help of cross-lingual bootstrapping.

## 7.5 Punjabi

Kumar [54] performed the comparison of HMM and Dynamic Time Wrap (DTW) for the recognition of Punjabi isolated words. HMM obtained 91.3% of accuracy and DTW accounted 94.0% of accuracy. Dua et al. [18] used HMM for implementing Punjabi STT system for connected words in classroom environment and open environment. The WER of 5.9% and 8.35% was observed for classroom and open ENVIRONMENT, respectively. Ghai and Singh [21] discussed in detail all the efforts made for ASR for Indo-Aryan Languages like Assamese, Hindi, Bengali, Oriya, Punjabi, Marathi, Sinhala, Urdu by various researchers. ASR research lack for Punjabi and more work was done on Punjabi isolated words. Lata and Arora [55] done the study of Punjabi tones with relation to orthographic characters and concluded that starting and middle position give falling tones. Rising tone was indicated when tonal character is at last position. Ghai and Singh [22] worked for recognizing continuous speech in Punjabi with the help of HMM. The word recognition accuracy was 94.32% and the sentence recognition found to be 82.18%. Lata and Arora [56] have done the analysis of/h/ sound in Punjabi with PRAAT as well as MATLAB software on Malwai, a dialect of Punjabi. Kaur and Singh [47] used Power Normalized Cepstral Coefficients (PNCC) on

connected words to build Punjabi ASR using HMM. WER in noise free and noisy environment obtained was 16.28% and 32.08%, respectively. Kaur and Singh [48] used MFCC, PLP, PNCC to build ASR for Punjabi connected words using HMM. WER obtained by MFCC, PLP and PNCC was noted as 13.19%, 16.28% and 16.28%, respectively, for noise free environment. Kumar and Singh [52] build the automatic spontaneous speech model using the Sphinx tool kit and Java programming. The model was tested for the corpus of Punjabi live interview speech. The recognition accuracy found to be 91.17% and 85.38% for Punjabi sentences and Punjabi words, respectively. Kadyan et al. [39] proposed heterogeneous feature extraction that had been used on the speech signal for neutralizing the effect of the various conditions of non-ideal environment. Various feature extraction techniques were used with hybrid HMM classifier like HMM with GA and HMM with Differential Evolution (DE). The Majhi obtained higher accuracy than Malwai, Powadhi and Doabi. The DE + HMM classifier gave the highest accuracy among the HMM, GA-HMM for both, speaker dependent and speaker independent, with MFCC. Dhanjal and Bhatia [16] developed text and speech corpus for the Malwai dialect of Punjabi. The corpus contained 200 sentences and 100 folk songs. Extended pronunciation feature was addressed. Kadyan et al. [40] evaluated the performance of ASR system for Punjabi using the combination of HMM with DE and with GA. The maximum WA in noisy environment was obtained by Malwai and it was 84.96%, 83.26% and 78.44% for DE-HMM, GA-HMM and HMM, respectively. Kumar and Singh [53] created automatic speaker independent speech model for recognizing the Punjabi speech. The model performance evaluated based on recognition accuracy for Punjabi words and Punjabi sentences. Guglani and Mishra [26] used the Kaldi ASR system for the recognition of Punjabi speech. MFCC gave less WER than PLP for Punjabi continuous speech. Kadyan et al. [41] reduced acoustic

**Table 6** Some of the Important Work on Indo-European Tonal Languages

| Author                      | Year  | Language   | Dataset                                                                                                    | Feature Extraction | Classifier                    | LM/ENV.                          | Result                                                                                                     |
|-----------------------------|-------|------------|------------------------------------------------------------------------------------------------------------|--------------------|-------------------------------|----------------------------------|------------------------------------------------------------------------------------------------------------|
| Vu et al. [126]             | 2010  | Croatian   | GP Corpus<br>Speaker: 100<br>Words: 100<br>sentences read<br>by each of the<br>speakers<br>Duration: 400 h | MFCC               | HMM                           |                                  | WER (%)<br>16.9 (Bulgarian)<br>32.8 (Croatian)<br>23.5 (Czech)<br>20.4 (Polish)<br>36.2 (Russian)          |
| Gulic et al. [27]           | 2011  | Croatian   | Speaker: 1<br>Words: 1031<br>Duration: 15 min                                                              | MFCC               | GMM-HMM                       |                                  | SER (%)    WER (%)<br>50        7.7                                                                        |
| Lu et al. [75]              | 2011  | Croatian   | GP Corpus<br>Speaker: 100<br>Words: 100<br>sentences read<br>by each of the<br>speakers<br>Duration: 400 h | MFCC               | GMM<br>SGMM                   |                                  | WER (%)<br>34.3<br>31.1                                                                                    |
| Dua et al. [17]             | 2012a | Punjabi    | Words: 30–35                                                                                               | MFCC               | HMM                           | ENV.<br>Classroom<br>Open        | Accuracy (%)<br>95.63<br>94.08                                                                             |
| Dua et al. [18]             | 2012b | Punjabi    | Words: 30–35                                                                                               | MFCC               | HMM                           | ENV.<br>Classroom<br>Open        | WER (%)<br>5.9<br>8.35                                                                                     |
| Ghai and Singh [23]         | 2013b | Punjabi    | Speakers: 3 males,<br>3 females<br>Words: 200                                                              | MFCC               | HMM                           | Whole word<br>Triphone           | Word Recognition<br>Accuracy (%)<br>Isolated    Con-<br>nected<br>92.05        87.75<br>97.14        91.62 |
| Rudzionis et al. [106]      | 2014  | Lithuanian | Diseases names: 217<br>Complaints: 208<br>Drugs names: 306                                                 | MFCC               | HMM                           | –                                | Accuracy (%)<br>98.92                                                                                      |
| Rasymas and Rudzionis [105] | 2015a | Lithuanian | Speakers: 7 males,<br>5 females<br>Diseases names: 25<br>Drugs names: 25                                   | MFCC               | KNN<br>LDA<br>QDA<br>LR<br>ML |                                  | Accuracy (%)<br>98.16<br>93.16<br>98.05<br>93.60<br>89.70                                                  |
| Rasymas and Rudzionis [104] | 2015b | Lithuanian | Speakers: 7 male, 5<br>female<br>Diseases names: 25<br>Drugs names: 25                                     | MFCC               | KNN<br>CART                   |                                  | Accuracy (%)<br>98.16<br>97.58                                                                             |
| Kaur and Singh [47]         | 2016a | Punjabi    | ENV: Clean<br>Speakers: 3<br>Words: 43<br>ENV: Noisy<br>Speakers: 3<br>Words: 53                           | PNCC               | HMM                           | ENV:<br>Noise free<br>ENV: Noisy | WER (%)<br>16.28<br>32.08                                                                                  |
| Kumar and Singh [51]        | 2016a | Punjabi    | Punjabi live speech<br>Sentences: 461<br>Words: 1213                                                       | MFCC               | HMM                           |                                  | Recognition accuracy<br>(%)<br>Sentences    Words<br>98.6        98.8                                      |
| Lileikyte et al. [70]       | 2016  | Lithuanian | Duration: 340 h<br>Untranscribed<br>speech                                                                 | BNF                | DNN-sMBR                      | –                                | WER (%)<br>18.3                                                                                            |

**Table 6** (continued)

| Author                   | Year  | Language   | Dataset                                                                                                                                                        | Feature Extraction         | Classifier                                         | LM/ENV.                                   | Result                                                                                                            |
|--------------------------|-------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------------------------------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Kadyan et al. [39]       | 2017a | Punjabi    | Speakers: 25<br>Words: 5000<br>Duration: 32 h                                                                                                                  | MFCC                       | HMM<br>GA-HMM<br>DE-HMM<br>HMM<br>GA-HMM<br>DE+HMM | Speaker Dependent<br>Speaker Independent  | Accuracy (%)<br>60.97<br>70.42<br>74.59<br>58.48<br>67.36<br>72.47                                                |
| Kadyan et al. [40]       | 2017b | Punjabi    | Training<br>Speakers: 24<br>Words: 58,700<br>Duration: 46 h<br>40 min<br>Testing<br>Speakers: 13<br>Words: 6100<br>Duration: 5 h<br>30 min                     | MFCC                       | DE-HMM<br>GA-HMM<br>HMM<br>DE-HMM<br>GA-HMM<br>HMM | Env: Noisy<br>Env: Real                   | WA (%)<br>84.96 (Malwai)<br>83.26 (Malwai)<br>78.44 (Malwai)<br>82.26 (Malwai)<br>78.47 (Majhi)<br>80.49 (Malwai) |
| Lileikyte et al. [71]    | 2017  | Lithuanian | FLP<br>Duration: 40 h<br>Transcribed speech<br>VLLP<br>Duration: 3 h                                                                                           |                            |                                                    | FLP<br>VLLP                               | WER (%)<br>42.4<br>52                                                                                             |
| Guglani and Mishra [26]  | 2018  | Punjabi    | Speakers: 24<br>Numbers: 0–9 digits<br>in Malwai dialect<br>Duration: 240 h                                                                                    | MFCC<br>PLP<br>MFCC<br>PLP | GMM-HMM                                            | Monophone<br>3-gram<br>Triphone<br>3-gram | WER (%)<br>31.0<br>33.1<br>21.2<br>22.7                                                                           |
| Kadyan et al. [41]       | 2018  | Punjabi    | Dataset1: Connected<br>speech corpus<br>Speakers: 13<br>Connected words:<br>21,764<br>Dataset2: Continuous<br>speech corpus<br>Sentences: 3611<br>Speakers: 13 | MFCC<br>MFCC<br>GFCC       | DNN-HMM                                            | –                                         | WER (%)<br>Dataset1<br>39.73<br>Dataset2<br>5.22<br>24.67                                                         |
| Safarik and Mateju [107] | 2018  | Latvian    | TEDx Speech<br>Durations: 2 h<br>Words: 13,514<br>Read Speech<br>Durations: 30 min<br>Words: 2880                                                              | MFCC                       | DNN                                                |                                           | WER (%)<br>27.78 (TEDx speech)<br>15.28 (Read speech)                                                             |
| Arora et al. [4]         | 2019  | Punjabi    | Punjabi dialect's<br>Majhi<br>Doabi<br>Malwai                                                                                                                  | –                          | –                                                  | –                                         | Mean pitch<br>370.23<br>162.23<br>194.190                                                                         |

mismatch between training and testing conditions with the help of DNN-HMM and handled the over fitting issue of training data with DNN-HMM. Arora et al. [4] studied pitch, intensity and fundamental frequency and their effect on Punjabi dialects.

## 8 Synthesis Analysis of Work Done for ASR Tonal Languages

Some of the common findings from the literature review has been presented initially and afterwards language wise review has also been given

- For performing various experiments, it is required that the standard benchmarks speech corpus is available especially in Indian language and therefore it presents a direction.
- Heterogeneous techniques of feature extraction neutralize the effect of conditions of non-ideal environment when used on input speech signal
- It is important to note here that for achieving good accuracy rates, the most important job of a researcher is to use hybrid feature extraction method. This methods supplies reliable and useful information to the classifier.
- For the recognition of tonal language, the important factor is the how accurately tonal features are being extracted.
- Combination of classifiers always depend upon the information supplied by each classifier.
- Because of too many variations in the speaking styles of individuals, speech recognition is undoubtedly a challenging task.
- Accuracy in speech recognition process always relies on features having high discriminating power. So, there is a dire need to study the feature selection algorithms for achieving good accuracy of under resourced languages.
- With different characteristics and variations in speech signals, and the distinct tones and dialects, it is challenge for a researcher to work on different speech dialects with good accuracy.
- As the working of the classifier for building acoustic model depends on the features extracted by the feature extraction phase, so it is mandatory to have a careful selection of the number of features for extraction methods and classifiers.
- Lack of benchmark speech corpus for majority languages hindered the research for novel innovations in developing speech recognitions in under resourced languages
- In most of the African languages like Yoruba, Hausa etc. very less literature is available.
- Accuracy of the system reduces with loud as well as with unpredictable noises.
- By increasing the number of speakers for speech corpus, surely there will be an increase in the recognition rate.
- Studies and work done in tonal languages like Chinese, Mandarin, Vietnamize are satisfactory with good results.
- All Indian tonal languages are not wee resourced. Languages like Punjabi, Dogri, Mizo and other Indo-tibetan languages lack resources for development of good ASR.
- It is observed that MFCC and HMM are most commonly used feature extraction and classification techniques.
- Studies in Croatian language have shown that better results can be obtained when phones are defined as syllable rather than individual letters.
- Word error rate in Swedish language is reduced with multilingual subspace
- It is also analyzed for the above findings that the HMM classifier is very common for the recognition tasks with good accuracy. But again, some findings proved that accuracy rates can be increased with HMM, ANN and some other hybrid techniques

Some of the specific findings of the languages have been explained below.

## 8.1 Chinese

- F0 as well as features of energy for neutral tone are not reliable discriminative features.
- The response competition problem may be exploited by the pitch anchoring hypothesis.
- Long segment of test speech limits the enhancement of pitch flux features.
- SDPBMM gives good result with small amount of dialectal speech data.
- GMBM proved better than Vector code book for Chinese dialect identification.
- The ISDP set approach proved to be flexible as well as faster approach than the syllable modeling for the Chinese speech recognition. The ISDP can be used for the applications having the computational power low.
- There is need to detect soft tone nuclei for speaker independent as well as to study the variability's of f0 of various speakers in speaker independent.
- Most of the studies have used Mel-cepstral features for the identification of Chinese dialect than the pitch-based features. The accuracy is increased as the size of the speech corpus is increased in MFCC, but the accuracy has not achieved tremendous gain in the case of Pitch flux features as the size of the speech corpus is increased.
- Full speech recognition or aligned transcription is not required for recognizing the tone with the help of vowel landmark detection.
- The falling-rising and falling tones of Chinese are easily recognized by the temporal cues than the tone flat tone and rising tone.
- Less calculation was required in the SFNNAM as compared to HMM Model and Chinese language Consonant Vowel (CV) structure can be easily implemented by the SFNNAM.

- For Mandarin, Bayes Decision Tree came out to be faster and accurate classifier than HMM and BN obtained higher rate than Broadcast Conversational (BC).

## 8.2 Thai

- Tonal scores can improve the performance of Thai spelling speech recognition system only if tonal weight is adjusted properly.
- The classification of vowel sounds is easier than initial and final consonants.
- Initial and final consonant classification is difficult as they have low energy and their occurrence is of short period.
- Final consonant has higher classification rate than initial consonants and vowel.
- Accuracy of continuous spelling can be improved with more study of recognition parameters.
- Better recognition performance can be obtained for classifying phonemes as well as tonal accents if the syllables are pronounced by the speakers in training set.
- Better recognition performance will be obtained if tested syllables are present in training set.
- The type of tone classification in Thai is not possible with the rising level of  $f_0$  but flattening, oscillation also help in the identification of tone in Thai language.
- Benchmark evaluation is very less for Thai LVCSR.
- Costly labeled training data used for tone learning as well as recognition can be reduced by intrinsic structure of tone.
- The recognition rate of the consonant recognition may be improved with the classified tone algorithm.

## 8.3 Vietnamese

- Studies have shown that better results can be obtained with MFCC than PLP
- The initial consonant of Vietnamese does not contain the information of tone as well as does not participate in the construction of tone of Vietnamese syllable.
- The initial consonant only creates the syllable.
- Only the final part of the syllable is affected by the Vietnamese tone.
- The CI can outperform the cross lingual CD approach with more hours of speech
- Flavor approach allowed the integration of various knowledge sources easily.
- Vietnamese speech recognition system can be improved by tonal features of phoneme.

## 8.4 Lithuanian

- Requirement of large size corpus and language model for increasing the accuracy of speech recognizer.
- LRN1 is useful for construction as well as evaluation of Lithuanian speech recognition which being independent and continuous.
- Better recognition performance can be obtained by modeling softness as well as accent information.
- Triphones outperform monophone in terms of recognition performance enriched with linguistic information.
- Derivation of triphone set from phone set is recommended to be used as modeling unit set.

## 8.5 Latvian

- The errors in speech recognition of Latvian are due to OOV and misrecognized word endings.
- No standard VLCSR is available.
- Recognition of adjectives and verbs is more difficult as compared to prepositions and particles.
- Problem is raised with non-canonical pronunciation.
- Very less reported research work

## 8.6 Punjabi

- Less work is done on the spontaneous and continuous speech.
- Most of the work is done on isolated or connected words.
- No standard speech corpus available.
- Work mostly done on the small size of vocabulary not on large size of vocabulary.
- Without incorporation tonal information size of vocabulary increases the error rate.
- Requirement of more speakers and vocabulary in Punjabi corpus for enhancing the recognition accuracy.

# 9 Technology Progress ASR

The intensively research is done in globally during the last three decades in the tonal language speech recognition, in the signal processing areas, SR algorithms, architectures and hardware. The summary of the technology progress for tonal Languages speech recognition is listed in the Table 7 based on past and present technologies. The summary of the technology progress for tonal languages SR is listed in the Table 7 based on past and present technologies.

**Table 7** ASR system progress on yearly basis

| Year          | Progress of tonal language SR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1952          | Digit recognizer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 1976          | Constrained grammar with 1000 word connected recognizer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 1980          | LSM recognizer with 1000 words                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 1988          | Typewriter phonetic in nature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 1993          | Read texts                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 1998          | Telephonic conversations<br>BN<br>From broadcast news speech retrieval                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 2002          | Controlled ENV<br>Limited tasks<br>Vocabulary large in size<br>Meetings related rich transcriptions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 2004          | Vocabulary unlimited in size based on morphemes<br>Dictation online in nature<br>Silencing Voice Recognition Technology (VRT) used the stick on sensors for grabbing the word before it is spoken but it does not senses the words exactly                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 2006          | Speech translation of broadcast                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 2008          | Arbitrary ENVIRONMENT<br>Vocabulary large in size<br>Limited tasks<br>Voice Recognition System (VRS) designed for visually impaired that used the voice command for accessing the internet. But does not gave the accurate result                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 2009          | SR adapt quickly the synthesized voice                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 2011          | Vocabulary unlimited<br>Tasks unlimited<br>Various languages<br>The use of multilingual systems for the multimodal devices which are speech enabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 2013          | Acoustic interference cancellation system that reduced the Smart TV's acoustic interference                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 2014          | Smart wheel chair used the command of touchpad as well as voice for moving the wheelchair                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 2015          | Encryption and decryption of the phone data through voice is done with the help of Voice Encrypted Recognition Authentication (VERA). VERA fails to access the data because of system failure                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 2016          | Laughing voice recognition senses the laughing periodic waveforms. But this system lack as it mixes various waveform easily<br>Automation processes was done with the wireless room automation system with the help of voice commands but it cannot do the automation process in the case of system failure<br>Smart mobile attendance system used the voice and fingerprint for the recognition<br>For the functioning of engine the voice commands used in the intelligent automatic starting engine but the system does not work in the system failure<br>Driving awareness detection evaluated awareness basis on answered of questions. But in this system evaluation of answer can be manipulated accordingly |
| 2017 and 2018 | Software called Dragon Professional Individual v15 used the concept of deep learning<br>Google Docs Voice Typing<br>Braina Pro<br>Winscribe SR<br>Cue-me                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

## 10 Issues and Challenges

Applications of speech recognition is significantly used in various commercial and governmental sections products like Alexa, Google speak, and Siri are some of the prominent products based on speech recognition. Speech is normally ingrained as a persuasive biometric modality and gained eloquent research interest. Voice as a biometric has also gained acceptance in the market. The area of speaker identification

deals with this. Various challenges for the speech recognition system are as follows:

- Most of the language around world lacks benchmark speech corpus
- Variation in speaking style among the speakers of different regions (of same language) exceed the variations inherent to every single speaker.

- Better accuracy rate can be achieved when large amount of speech data is processed and managed.
- To increase and achieve the high recognition rate variety of speech, tone and emotion embedded speech corpus is very crucial
- To recognize an unknown sample of speech data from a large corpus of known speakers with high confidence.
- To receive necessary speech data required for obtaining reliable distribution estimates is itself a challenging issue.
- To acquire deep understanding of the languages and its characteristics.
- To have a survey of the already completed recognized work in the speech recognition of various under-resourced languages.
- To gain knowledge of already existing approaches and to create a novel approach may be hybrid for the feature extraction and classification methods.

## 11 Inferences and Future Directions

The ASR input data in the form of audio and generate the output in the form of symbol in specific order with respect to the order of the words spoken during input speech. For the Asian, African, Indo-European tonal languages the comprehensive survey related to ASR system with the various features, classifiers as well as the accuracy is presented. It helps the reader to get the idea regarding the nature of the study, features as well as classification methods. The user obtained the information that how the accuracy is enhanced by the size of the dataset. From the survey we also try to present that in terms of different accuracy rates, research in the area of speech recognition for different languages spoken around

the globe. The paper presented a collection of good findings with acceptable accuracy in languages such as Arabic, Chinese, and Thai. The lack of a standard corpus on various African and Indo-European languages are also the major concerns in this respect. High accuracy rates, research filed in ASR in the tonal language of Asian continent is presented. The ASR system for tonal languages of the American and the Austral-Asia are not studied. There is lack of work for the, African, Indo-European tonal language ASR. Synthesis analysis as well as the comparative study with respect to the reported work has been mentioned. Through paper, we also present list of features and classifiers for the speech recognition system for tonal languages in summarized form. For the future directions, it can be contemplated that there is a need to develop novel approaches for the feature extraction, classification methods and hybrid approaches for both features and classifiers for achieving the best accuracy results. The problem of maintaining corpus with enough speakers can also be a novel future aspect in this direction. In the future the ASR system for American and Austral-Asia tonal languages will be reviewed as well as the tonal languages of Asia, African and Indo-European that are not reviewed will be studied in the future. The development towards the new feature extraction methods with more maturity levels and the accuracies will be an important issue for the researchers. Also, more promising research on the speech to speech translation are novel subtopics in this direction.

## Compliance with Ethical Standards

**Conflict of interest** On behalf of all authors, the corresponding author states that there is no conflict of interest.

## Appendix 1: A Quality Assessment Forms

### Screening Question

#### Section-1

Does the research paper refer to SR?

Yes

Consider:

This paper contains the study of SR. Case study, experimental study or the research paper are included

Section1- is evaluated first. If the positive reply is received then proceed to Section 2.

#### Screening question

#### Section-2

Key Sub –Area Categorization

Has the research paper focused on SR of tonal language?

Yes

Consider

Is the study's focus or main focus is on SR of tonal language or not?

Did the study fitted in the above mentioned subareas categorized?

If the study's vital aim is on is on SR then proceed to Section3, else proceed to Section-4

#### Detailed questions

#### Section-3

##### Findings

Is there a clear statement of findings?

Yes

Consider

Did the study mention the approach/SR of tonal language

Has the SR technique reported?

What is the corresponding transformation technique, finding, i.e. source representation

##### Comparison

Is there sufficient data for comparative analysis?

Yes

Consider

Is there discussion regarding the important parameters done

Is the study referring to tonal language SR explicitly?

#### Detailed questions

#### Section-4

##### Findings

Has study mentioned the type of SR of tonal language?

Yes

Consider

How well the SR of tonal language is categorized?

Has the study explicitly mention the type of SR of tonal language, or has to be inferred from the study.

## Appendix 2: Data items Extracted from All Papers

| Data item                                    | Description                                                                                                                         |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Token Initialization                         | Unique Identifier for the study                                                                                                     |
| Data source                                  | Author, title year                                                                                                                  |
| Category of article                          | Article from journal, conference, workshop.                                                                                         |
| Study objectives                             | What are the objectives of the study, i.e., search focus, i.e., the research areas of the research on which the paper focuses.      |
| Design of study                              | Category of study—feature extraction, classification, SR, comparative analysis, etc.                                                |
| What is the SR technique of tonal languages? | Spoken words are transcribed by the computers into readable text.                                                                   |
| Method of comparison                         | Important parameters values for SR, i.e., f0, tone, energy, gain, amplitude, power, frequency response, voiced and unvoiced feature |
| Data analysis phase                          | Data analysis with respect to presentation of source, error rate, recognition accuracy                                              |
| Tool and the usage                           | It refers to the SR tool, developer and usage of the tool                                                                           |
| Result of study                              | Result or conclusions from the primary study                                                                                        |

## Appendix 3

|       |                                       |
|-------|---------------------------------------|
| A     | Tonal Assimilation                    |
| ACC   | Letter Accuracy                       |
| AI    | Artificial Intelligence               |
| AMDF  | Average Magnitude Difference Function |
| ANN   | Artificial Neural Network             |
| ASR   | Automatic Speech Recognition          |
| ATWV  | Actual Term Weighted Value            |
| BC    | Broadcast Conversational              |
| BCC   | Broadcasting Corporation of Chinese   |
| BLSTM | Bidirectional Long Short Term Memory  |
| BN    | Broadcast News                        |
| BTEC  | Basic Travel Expression Corpus        |
| CAP   | Computer Assisted Pronunciation       |

|          |                                                             |
|----------|-------------------------------------------------------------|
| CART     | Classification And Regression Tree                          |
| CD       | Context Dependent                                           |
| CD-T-175 | Context Dependent Model                                     |
| CE       | Consonantal Effects                                         |
| CER      | Character Error Rate                                        |
| CI       | Context Independent                                         |
| CI-T-5   | Context Independent Model                                   |
| CNN      | Convolutional Neural Network                                |
| COR      | Letter Correctness                                          |
| CPM      | Compound Pseudo-Morpheme                                    |
| CRF      | Conditional Random Field                                    |
| CTC      | Connectionist Temporal Classification                       |
| CTS      | Conversational Telephone Speech                             |
| CV       | Consonant Vowel                                             |
| D        | Declination Normalization                                   |
| DBNFs    | Deep Bottle Neck Features                                   |
| DCTC     | Discrete Cosine Transform Coefficient                       |
| DE       | Differential Evolution                                      |
| DNN      | Deep Neural Network                                         |
| DR       | Detection Rate                                              |
| DRT      | Diagnostic Rhyme Test                                       |
| DTW      | Dynamic Time Wrap                                           |
| EMG      | Electromyography                                            |
| ENV      | Environment                                                 |
| F        | Average Pitch Value of Whole Utterance                      |
| f0       | Fundamental Frequency                                       |
| F0       | Time Periods of the Successive Human Vocal Cords Vibrations |
| FFT      | Fast Fourier Transform                                      |
| fi       | Average Pitch Values of Voiced Segment                      |
| FLP      | Full Language Pack                                          |
| G2P      | Grapheme-to-Phoneme                                         |
| GA       | Genetic Algorithm                                           |
| GFCC     | Gammatone Frequency Cepstral Coefficient                    |
| GMBM     | Gaussian Mixture Bigram Model                               |
| GMM      | Gaussian Mixture Model                                      |
| GP       | Global Phone                                                |
| HCRF     | Hidden Conditional Random Field                             |
| HINT     | Hearing In Noise Test                                       |
| HMM      | Hidden Markov Model                                         |
| H-T-30   | Half-Tone Model                                             |
| I2R      | Institute for Infocomm Research                             |

|          |                                                              |                      |                                                                                                       |
|----------|--------------------------------------------------------------|----------------------|-------------------------------------------------------------------------------------------------------|
| IC       | InterCorp                                                    | MLLT                 | Maximum Likelihood Linear Transformation                                                              |
| IF       | Initial/Final                                                |                      |                                                                                                       |
| ISDPs    | Intra Syllable Dependent Phone Set                           | MLP                  | Multi-Layer Perceptron                                                                                |
|          |                                                              | MMI                  | Maximum Mutual Information                                                                            |
| KLT      | Karhunen–Loeve Transformation                                | MSD                  | MultiSpace Distribution                                                                               |
|          |                                                              | MT                   | Machine Translation                                                                                   |
| KNN      | Nearest Neighbors                                            | NCC                  | Normalized Cross-Correlation                                                                          |
| KWS      | Keyword Search                                               | NECTEC               | National Electronics and Computer Technology Center                                                   |
| KWS      | Keyword Spotting                                             |                      |                                                                                                       |
| LCR      | Letter Correct Rate                                          | NECTEC-ATR           | National Electronics and Computer Technology Center                                                   |
| LDA      | Linear Discriminant Analysis                                 |                      |                                                                                                       |
| LM       | Language Model                                               |                      | Advanced Telecommunications Research                                                                  |
| LM_DP_MI | Language Model with Dynamic Programming                      | NN                   | Neural Network                                                                                        |
|          | Word Segmentation Method                                     | Norm_Log_F0_Mean_Dev | f0 Logarithmic Value and normalizing the value of f0 by mean and standard deviation of every sentence |
| LM_MM    | Language Model with Polysyllabic Words with Maximum Matching |                      |                                                                                                       |
| LM1      | Closed Type Model                                            | OOV                  | Out of Vocabulary                                                                                     |
| LM2      | Mix Type Model                                               | ORCHID               | Open Linguistic Resources                                                                             |
| LM3      | Open Type Model                                              |                      | Channeled Toward Interdisciplinary Research                                                           |
| LOTUS    | Large Vocabulary Thai Continuous Speech                      | PB                   | Phonetically Balanced                                                                                 |
|          |                                                              | PCR                  | Phoneme Correct Rate                                                                                  |
| LPCC     | Linear Predictive Cepstral Coefficient                       | PD                   | Phonetically Distributed                                                                              |
|          |                                                              | PER                  | Phone Error Rate                                                                                      |
| LPM      | Latent Prosody Model                                         | PLP                  | Perceptual Linear Prediction                                                                          |
| LR       | Logistic Regression                                          | PLP                  | Perceptual Linear Predictive                                                                          |
| LRN 0.1  | Lithuanian Radio News Prototype Version 0.1                  | PM                   | Pseudo-Morpheme                                                                                       |
|          |                                                              | PNCC                 | Power Normalized Cepstral Coefficients                                                                |
| LRN1     | Lithuanian Radio News Version 1                              |                      |                                                                                                       |
| LSTM     | Long Short-Term Memory                                       | QDA                  | Quadratic Discriminant Analysis                                                                       |
| LSTM-OP  | Long Short-Term Memory with Output Projection                | RLAT                 | Rapid Language Adaptation Toolkit                                                                     |
|          |                                                              | RNN                  | Recurrent Neural Network                                                                              |
| LSTMP    | Long Short-Term Memory Projected                             | SA                   | Syllable Accuracy                                                                                     |
| LVCSR    | Large Vocabulary Continuous Speech Recognition               | SBN                  | Stacked Bottle Neck                                                                                   |
|          |                                                              | SD                   | Spoken Documents                                                                                      |
| LVQ      | Learning Vector Quantization                                 | SDPBMM               | State-Dependent Phoneme Based Model Merging                                                           |
| MBN      | Mandarin Broadcast News                                      |                      |                                                                                                       |
| MCE      | Minimum Classification Error                                 | SER                  | Sentence Error Rate                                                                                   |
| MFCC     | Mel-Frequency Cepstral Coefficient                           | SFNNAM               | State Feedback Neural Network Activation Model                                                        |
|          |                                                              | SGMM                 | Subspace Gaussian Mixture Model                                                                       |
| MFPLP    | Mel Frequency with Perceptual Linear Prediction              |                      |                                                                                                       |
| MGCPM    | Mixed Gaussian Continuous Probability Model                  | SLER                 | Syllable Error Rate                                                                                   |
|          |                                                              | SLM                  | Syllable Lattice Matching                                                                             |
| ML       | Maximum Likelihood                                           | sMBR                 | State-Level Minimum Bayes Risk                                                                        |
| MLE      | Maximum Likelihood Estimation                                |                      |                                                                                                       |
| MLLR     | Maximum Likelihood Linear Adaption                           | SQ                   | Speech Queries                                                                                        |
|          |                                                              | SR                   | Speech Recognition                                                                                    |
|          |                                                              | SRU                  | Speech Recognition Unit                                                                               |

|        |                                            |
|--------|--------------------------------------------|
| STT    | Speech-To-Text                             |
| SVM    | Support Vector Machine                     |
| TAR    | Tone Accuracy Rate                         |
| TCR    | Tone Correct Rate                          |
| TD     | Text Documents                             |
| TDRT-I | Thai Diagnostic Rhyme Test for Initials    |
| TF     | Tone Features                              |
| Tone 2 | High-Rising                                |
| Tone 3 | Dipping                                    |
| Tone 4 | Falling                                    |
| Tone 5 | Undefined                                  |
| Tone1  | High-Level                                 |
| TQ     | Text Queries                               |
| VERA   | Voice Encrypted Recognition Authentication |
| VLLP   | Very Limited Language Pack                 |
| VNBN   | Vietnamese Broadcast News                  |
| VOV    | Voice of Vietnamese                        |
| VRS    | Voice Recognition System                   |
| VRT    | Voice Recognition Technology               |
| VSM    | Vector Space Model                         |
| WA     | Word Accuracy                              |
| WDC    | Wu Dialect Chinese                         |
| WER    | Word Error Rate                            |
| WFST   | Weighted Finite State Transducer           |
| WU     | Word Unit                                  |
| XIF    | Extended Initial/Final                     |

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